INTE 22283 – Mobile Application Development

1 - Introduction to Mobile Computing

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Overview

- Definition of Mobile Computing
- Mobility of Computers
- Mobile Eco System
- Significance of Mobile Applications
- Limitations & Issues
- Evolution of Mobile Devices
- Anatomy of a Mobile Device

Definition of Mobile Computing

- Mobile Computing is a technology that allows transmission of data (text, voice, video etc.) via a computer or any other wireless enabled device without having to be connected to a fixed physical link.

- The ability to use technology in remote or **mobile** (non static) environments.

- This technology is based on the use of battery powered, portable, and wireless **computing** and communication devices, like smart **mobile** phones, wearable computers and personal digital assistants (PDAs)

Mobility of Computers

- There are many areas related to the mobility in mobile Computing. This refers to mobility of different components of a mobile computing systems & users.

  - User Mobility
  - Network Mobility
  - Bearer Mobility
  - Device Mobility
  - Session Mobility
  - Service Mobility
  - Host Mobility

Mobile Ecosystem

- Mobile Ecosystem is a collection of
  - Multiple Devices (mobile phones, tablets etc.)
  - Software (operating system, development tools, testing tools etc.)
  - Companies (device Manufacturers, mobile carriers, app developers, app stores).
  - Users.
- It is the process of transferring data from one device to another or by device itself based on some programmatically instructions.
- For the sustainability of mobile eco system all parties must provide their proper contribution.

Mobile Ecosystem

Significance of Mobile Applications

- Mobile applications can be developed in order to provide highly beneficial features to all the parties not just for end users.
  - Ex:
    - Can be used to collect data for research purpose related to user mobility.
    - Can be used to identify user behavioral patterns for develop targeted marketing campaigns.

Limitations & Issues

- Deficient Bandwidth
  - Mobile web access is usually slower than direct cable connections
  - Bandwidth utilization is often improved and compression of information before transmission.
  - There can be frequent disconnections since the user mobility.
- Power consumption
  - Mobile Computers can deem their batteries because the primary power supply.
  - Batteries should be ideally as lightweight as potential however at constant time they must be capable of longer operation times.
  - Power consumption should be decreased to extend battery life.
  - Chips are redesigned to control at lower voltages.
  - Individual Components, should be high-powered down once they are idle.

Limitations & Issues

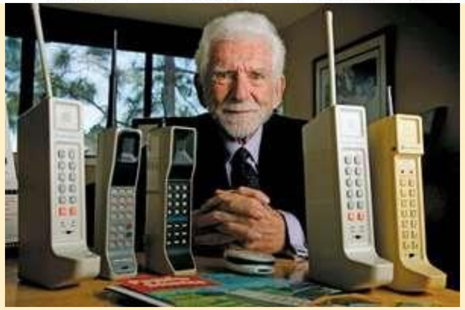
- Security Standards
  - Security could be a major concern whereas regarding the mobile computing standards on the fleet.
  - One will simply attack the VPN through an enormous variety of networks interconnected through the road
  - **Privacy** : Avoiding unauthorized users from fast access to essential data of any explicit customer.
  - **Reliability** : Ensures unauthorized modification, damage or creation of knowledge cannot occur.
  - **Availability**: Making certain approved users obtaining the access they need.
  - **Legitimate**: Making certain that only approved users have access to services.
  - **Accountability**: Making certain that the users are command liable for their security connected activities by composing the user and his/her activities are joined if and once necessary.

Limitations & Issues

- Software Platforms
  - Applications depend heavily on the operating system of the mobile device.
  - Cross Platform operability could not be achieved since one application developed for one mobile platform may not work on another

When developing mobile applications we need to think of a way to address all the issues at least to a certain level.

Evolution of Mobile Devices



Evolution of Mobile Devices

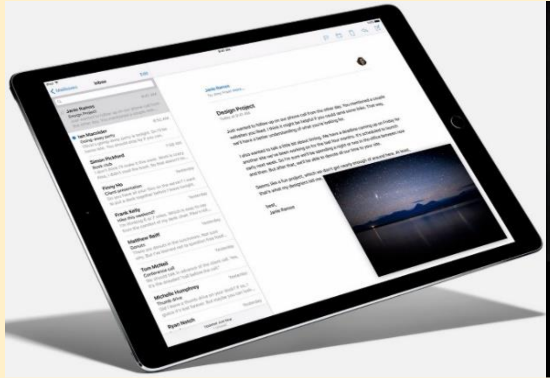
- The Conventional phone has changed from a Brick type to Touch type smart phone that we are using now together with their functionalities.

  - 1^{st} Gen – Brick Era (1973-1988)
  - 2^{nd} Gen – Candy Bar Era (1988 - 1998)
  - 3^{rd} Gen – Feature Phone Era (1998 - 2008)
  - 4^{th} Gen – Smart Phone Era
  - 5^{th} Gen – Touch Phone Era

Evolution of Mobile Devices

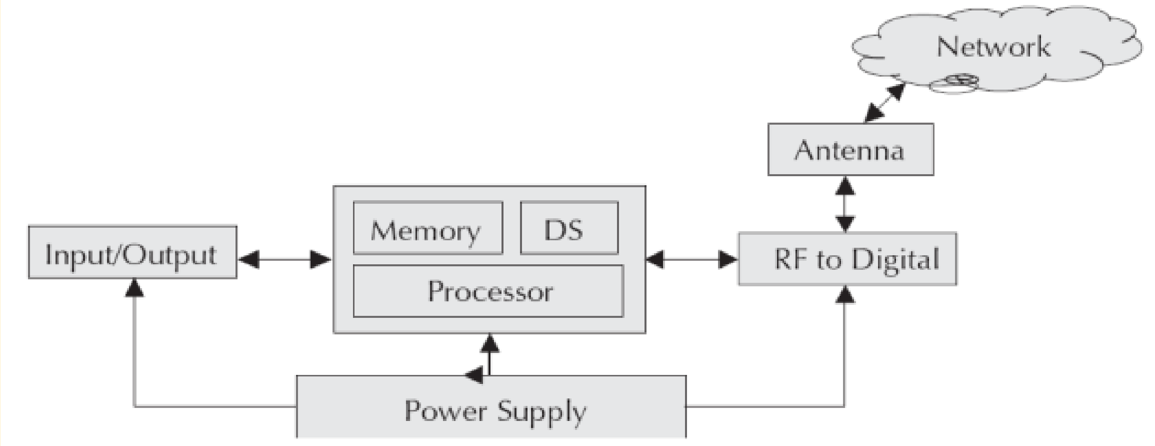
- It is not just Mobile Phones!

  - Tablet PCs
  - Smart Wear (Wearable Devices)
  - Robots
  - Smart Vehicles
  - And Many more…



Anatomy of a Mobile Device

- Microprocessor
- I/O Unit
- Antenna, Digital Signal Processor
- Power Source
- Memory (Persistent (ROM) & Volatile (RAM))



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